

Shark Colony Experiment

Shared-ocean ecology (GA + RL) — 2026-06-12T04:10:16

Summary

Summary	
Cycles requested / completed	1000 / 1000
Founding pod	30
Carrying capacity	100
Peak population	100
Final population	97
Colony went extinct?	no
Best fitness (any cycle)	137.400
Final mean fitness	-10.015
Runtime	52.7 s

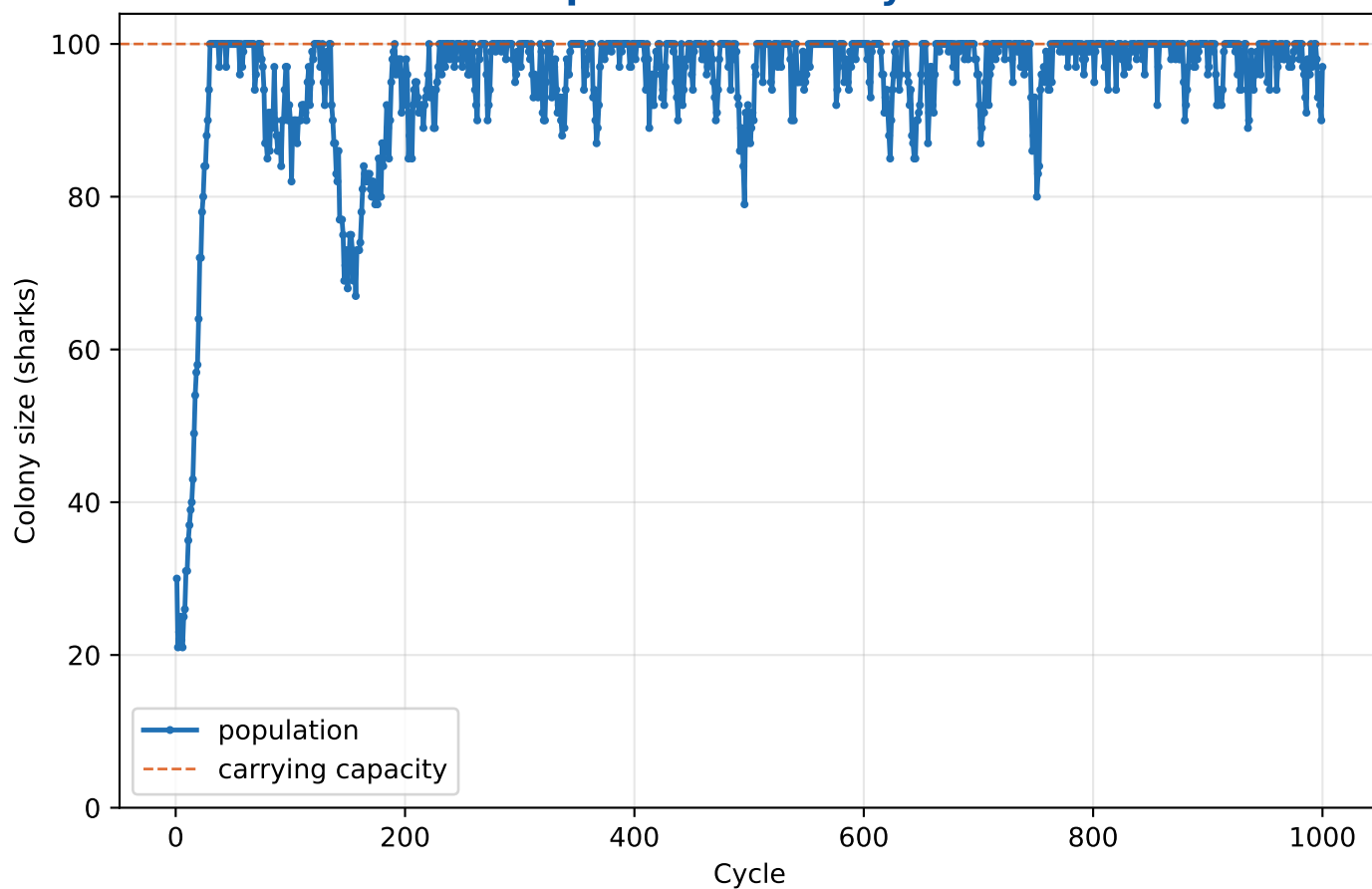
Ecology & competition

Ecology setting	Value
Shared, depleting ocean	True
Fish per shark / pool cap	1.8 / 30
Must eat to breed	True
Fitness-weighted breeding	True
Cull weakest on overcrowding	True
Gestation divisor / brood	6.0 / 2
Steps per life / brain warmup	80 / 400
Mutation rate / seed	0.2 / 0
Run timestamp	2026-06-12T04:10:16

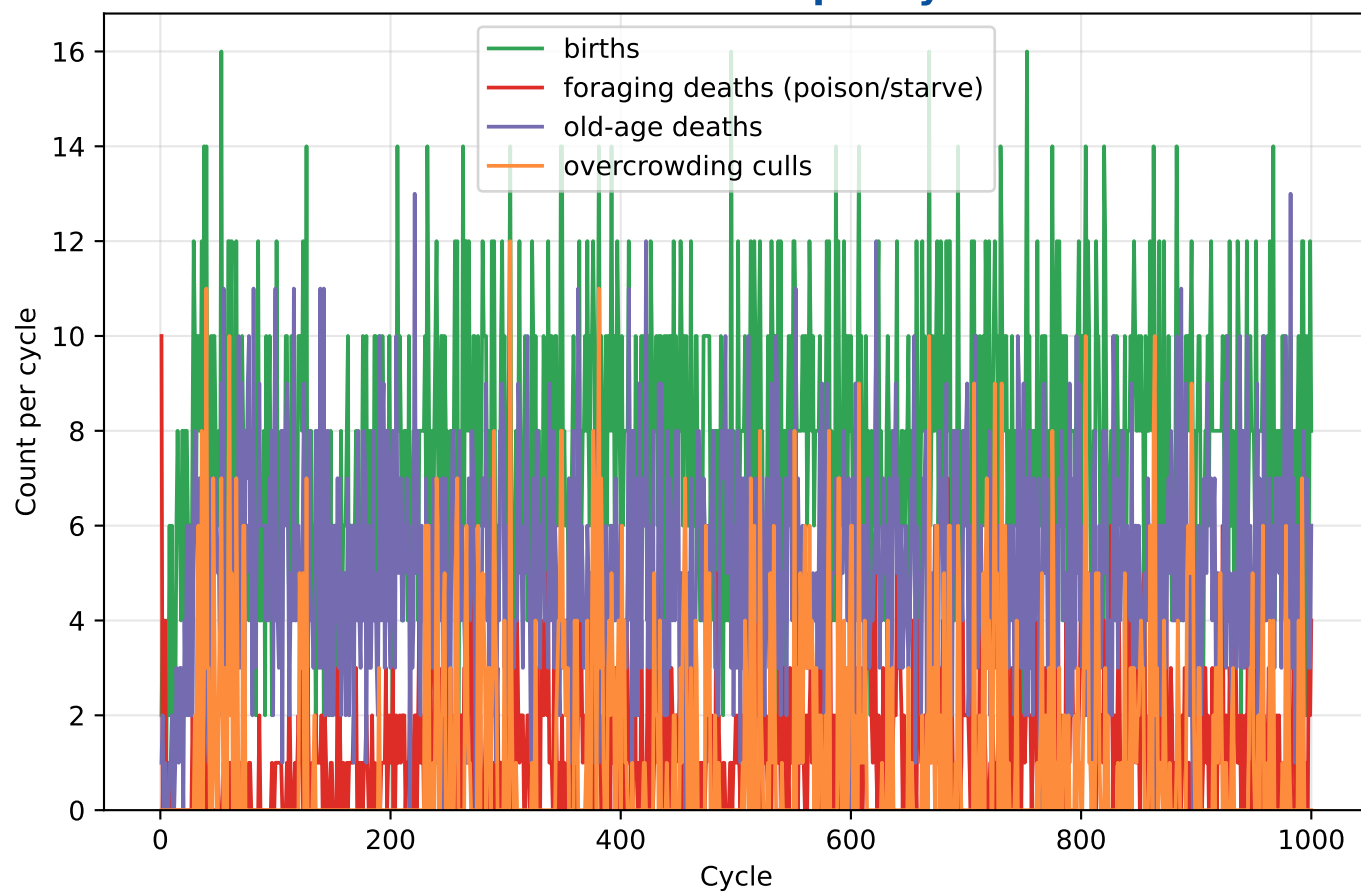
Shared brain (Q-learning)

Shared brain (Q-learning)	Value
States learned	236
Final epsilon (explore rate)	0.050
Epsilon min / decay	0.050 / 0.9950
Alpha (learning rate)	0.100
Gamma (discount)	0.950

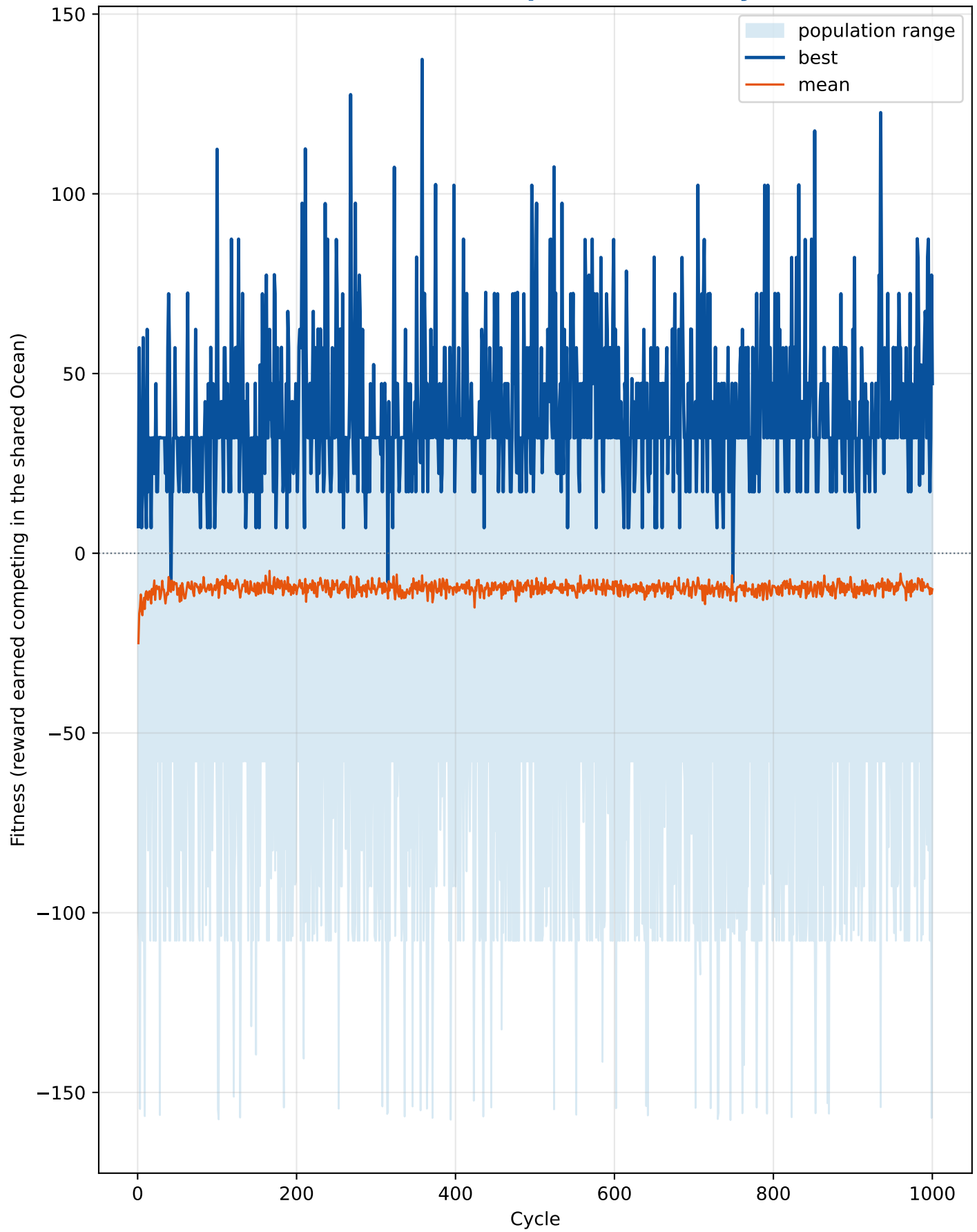
Population over cycles



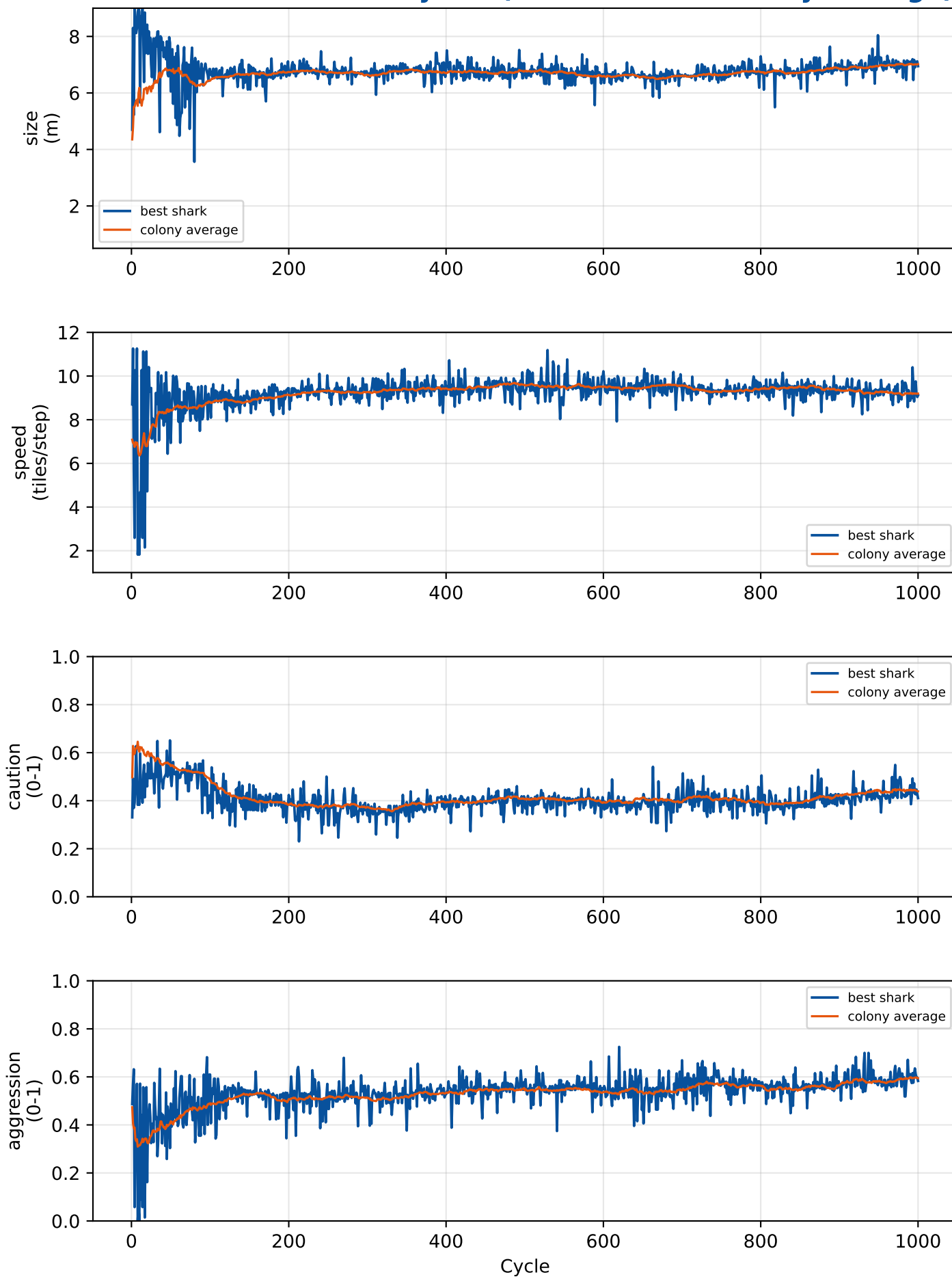
Births & deaths per cycle



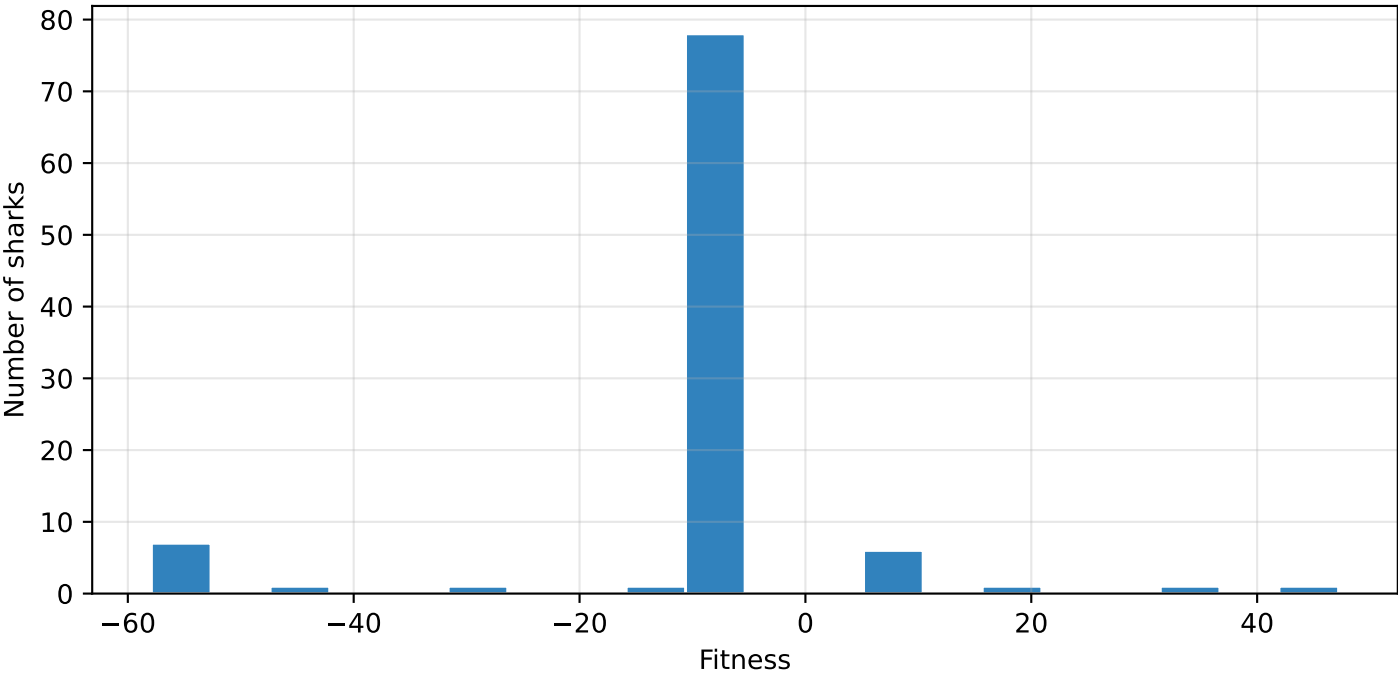
Fitness under competition over cycles



Evolvable traits over cycles (best shark vs. colony average)



Final-cycle fitness distribution (cycle 1000)



Best genome found (highest single-cycle reward under competition)

Trait	Value	Unit	Evolved?
size	6.808	m	yes
speed	9.164	tiles/step	yes
caution	0.331	0-1	yes
gestation_period	12.000	months	no (fixed)
aggression	0.509	0-1	yes
field_of_perception	5.000	tiles (radius)	no (fixed)
color_hue	205.000	degrees (HSV hue)	no (fixed)

Per-cycle summary (showing 31 of 1000 cycles, every 34)

Cycle	Pop	Size	Speed	Cautious	Aggression	Best	Mean	Born	Died	Cull
1	30	4.70	8.70	0.33	0.49	7.4	-25.0	2	11	0
35	100	6.68	7.05	0.37	0.47	22.2	-14.0	8	7	1
69	94	6.93	9.23	0.47	0.54	32.2	-9.4	6	4	0
103	90	6.57	9.19	0.33	0.55	32.2	-10.5	4	4	0
137	90	6.90	8.48	0.48	0.54	17.1	-9.0	6	9	0
171	80	5.70	9.21	0.42	0.47	17.1	-7.9	6	4	0
205	91	7.09	9.44	0.45	0.50	62.3	-9.2	6	12	0
239	98	6.80	10.10	0.36	0.55	87.4	-7.5	10	6	2
273	92	6.72	9.04	0.38	0.52	37.3	-12.1	10	8	0
307	100	6.68	9.90	0.34	0.54	47.2	-8.4	2	3	0
341	98	6.94	9.29	0.38	0.53	17.1	-11.1	6	8	0
375	99	6.88	9.31	0.38	0.51	102.6	-8.2	10	5	4
409	98	6.58	9.91	0.40	0.55	32.1	-9.9	4	8	0
443	94	6.88	8.88	0.39	0.61	57.2	-9.7	10	7	0
477	100	6.72	9.59	0.39	0.54	22.4	-11.5	10	5	5
511	100	6.80	9.45	0.40	0.48	32.1	-9.8	8	13	0
545	95	6.67	8.03	0.40	0.57	72.2	-9.1	8	7	0
579	100	7.20	9.17	0.41	0.55	32.2	-9.5	12	12	0
613	100	6.70	9.36	0.37	0.52	47.3	-10.7	10	8	2
647	90	6.56	9.48	0.36	0.63	32.1	-9.1	6	5	0
681	95	6.56	9.11	0.39	0.55	62.3	-6.8	10	0	5
715	98	6.60	9.52	0.41	0.61	32.1	-9.0	10	3	5
749	93	6.73	9.36	0.35	0.56	-8.0	-11.6	2	8	0
783	100	6.54	9.32	0.46	0.57	32.1	-11.8	10	10	0
817	100	6.74	9.41	0.37	0.51	32.2	-7.7	4	5	0
851	100	6.15	8.89	0.40	0.55	32.2	-9.6	8	5	3
885	100	7.00	9.22	0.39	0.56	72.2	-7.9	8	6	2
919	100	6.89	9.14	0.42	0.61	22.2	-8.8	8	7	1
953	98	7.25	9.82	0.35	0.60	32.2	-9.2	6	10	0
987	98	7.12	8.58	0.43	0.67	37.3	-7.2	4	6	0
1000	97	6.95	9.12	0.41	0.58	47.2	-10.0	8	10	0